[INICIAL] TCArg 2026 Contest #7

A. Cutting Carrot

2 seconds, 256 megabytes

Igor the analyst has adopted n little bunnies. As we all know, bunnies love carrots. Thus, Igor has bought a carrot to be shared between his bunnies. Igor wants to treat all the bunnies equally, and thus he wants to cut the carrot into n pieces of equal area.

Formally, the carrot can be viewed as an isosceles triangle with base length equal to 1 and height equal to h. Igor wants to make $n - 1$ cuts **parallel to the base** to cut the carrot into n pieces. He wants to make sure that all n pieces have the same area. Can you help Igor determine where to cut the carrot so that each piece have equal area?

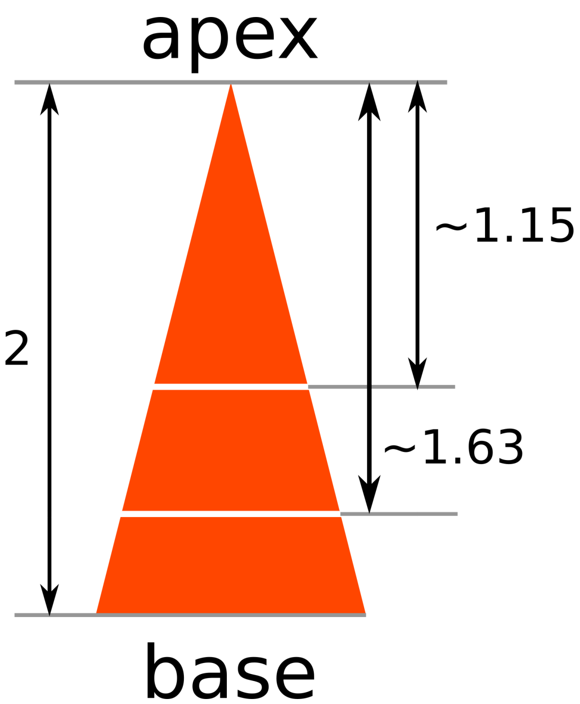

Illustration to the first example.

Input

The first and only line of input contains two space-separated integers, n and h ($2 \le n \le 1000$, $1 \le h \le 10^5$).

Output

The output should contain $n - 1$ real numbers $x_1, x_2, ..., x_{n-1}$. The number x_i denotes that the i-th cut must be made x_i units away from the apex of the carrot. In addition, $0 < x_1 < x_2 < ... < x_{n-1} < h$ must hold.

Your output will be considered correct if absolute or relative error of every number in your output doesn't exceed 10^{-6}.

Formally, let your answer be a, and the jury's answer be b. Your answer is considered correct if $\frac{|a-b|}{\max(1,b)} \le 10^{-6}$.

input
```
3 2
```
output
```
1.154700538379 1.632993161855
```

input
```
2 100000
```
output
```
70710.678118654752
```

Definition of isosceles triangle: https://en.wikipedia.org/wiki/Isosceles_triangle.

B. Game

2 seconds, 256 megabytes

Two players play a game.

Initially there are n integers $a_1, a_2, \ldots, a_n$ written on the board. Each turn a player selects one number and erases it from the board. This continues until there is only one number left on the board, i. e. $n - 1$ turns are made. The first player makes the first move, then players alternate turns.

The first player wants to minimize the last number that would be left on the board, while the second player wants to maximize it.

You want to know what number will be left on the board after $n - 1$ turns if both players make optimal moves.

Input

The first line contains one integer n ($1 \le n \le 1000$) — the number of numbers on the board.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \le a_i \le 10^6$).

Output

Print one number that will be left on the board.

input
```
3
2 1 3
```
output
```
2
```

input
```
3
2 2 2
```
output
```
2
```

In the first sample, the first player erases 3 and the second erases 1. 2 is left on the board.

In the second sample, 2 is left on the board regardless of the actions of the players.

C. Andrew and Taxi

2 seconds, 256 megabytes

Andrew prefers taxi to other means of transport, but recently most taxi drivers have been acting inappropriately. In order to earn more money, taxi drivers started to drive in circles. Roads in Andrew's city are one-way, and people are not necessary able to travel from one part to another, but it pales in comparison to insidious taxi drivers.

The mayor of the city decided to change the direction of certain roads so that the taxi drivers wouldn't be able to increase the cost of the trip endlessly. More formally, if the taxi driver is on a certain crossroads, they wouldn't be able to reach it again if he performs a nonzero trip.

Traffic controllers are needed in order to change the direction the road goes. For every road it is known how many traffic controllers are needed to change the direction of the road to the opposite one. It is allowed to change the directions of roads one by one, meaning that each traffic controller can participate in reversing two or more roads.

You need to calculate the minimum number of traffic controllers that you need to hire to perform the task and the list of the roads that need to be reversed.

Input

The first line contains two integers n and m ($2 \le n \le 100\,000$, $1 \le m \le 100\,000$) — the number of crossroads and the number of roads in the city, respectively.

Each of the following m lines contain three integers u_i, v_i and c_i ($1 \le u_i, v_i \le n$, $1 \le c_i \le 10^9$, $u_i \ne v_i$) — the crossroads the road starts at, the crossroads the road ends at and the number of traffic controllers required to reverse this road.

Output

In the first line output two integers the minimal amount of traffic controllers required to complete the task and amount of roads k which should be reversed. k should not be minimized.

In the next line output k integers separated by spaces — numbers of roads, the directions of which should be reversed. The roads are numerated from 1 in the order they are written in the input. If there are many solutions, print any of them.

input
```
5 6
2 1 1
5 2 6
2 3 2
3 4 3
4 5 5
1 5 4
```
output
```
2 2
1 3
```

input

```
5 7
2 1 5
3 2 3
1 3 3
2 4 1
4 3 5
5 4 1
1 5 3
```

output

```
3 3
3 4 7
```

There are two simple cycles in the first example: $1 \to 5 \to 2 \to 1$ and $2 \to 3 \to 4 \to 5 \to 2$. One traffic controller can only reverse the road $2 \to 1$ and he can't destroy the second cycle by himself. Two traffic controllers can reverse roads $2 \to 1$ and $2 \to 3$ which would satisfy the condition.

In the second example one traffic controller can't destroy the cycle $1 \to 3 \to 2 \to 1$. With the help of three controllers we can, for example, reverse roads $1 \to 3$,$2 \to 4$, $1 \to 5$.

D. Bear and Prime 100

1 second, 256 megabytes

This is an interactive problem. In the output section below you will see the information about flushing the output.

Bear Limak thinks of some hidden number — an integer from interval $[2, 100]$. Your task is to say if the hidden number is prime or composite.

Integer $x > 1$ is called prime if it has exactly two distinct divisors, 1 and x. If integer $x > 1$ is not prime, it's called composite.

You can ask up to 20 queries about divisors of the hidden number. In each query you should print an integer from interval $[2, 100]$. The system will answer "`yes`" if your integer is a divisor of the hidden number. Otherwise, the answer will be "`no`".

For example, if the hidden number is 14 then the system will answer "`yes`" only if you print 2, 7 or 14.

When you are done asking queries, print "`prime`" or "`composite`" and terminate your program.

You will get the `Wrong Answer` verdict if you ask more than 20 queries, or if you print an integer not from the range $[2, 100]$. Also, you will get the `Wrong Answer` verdict if the printed answer isn't correct.

You will get the `Idleness Limit Exceeded` verdict if you don't print anything (but you should) or if you forget about flushing the output (more info below).

Input

After each query you should read one string from the input. It will be "`yes`" if the printed integer is a divisor of the hidden number, and "`no`" otherwise.

Output

Up to 20 times you can ask a query — print an integer from interval $[2, 100]$ in one line. You have to both print the end-of-line character and `flush` the output. After flushing you should read a response from the input.

In any moment you can print the answer "`prime`" or "`composite`" (without the quotes). After that, flush the output and terminate your program.

To flush you can use (just after printing an integer and end-of-line):

- `fflush(stdout)` in C++;
- `System.out.flush()` in Java;
- `stdout.flush()` in Python;
- `flush(output)` in Pascal;
- See the documentation for other languages.

Hacking. To hack someone, as the input you should print the hidden number — one integer from the interval $[2, 100]$. Of course, his/her solution won't be able to read the hidden number from the input.

input

```
yes
no
yes
```

output

```
2
80
5
composite
```

input
```
no
yes
no
no
no
```
output
```
58
59
78
78
2
prime
```

The hidden number in the first query is 30. In a table below you can see a better form of the provided example of the communication process.

solution	system
2	
	yes
80	
	no
5	
	yes
composite	

The hidden number is divisible by both 2 and 5. Thus, it must be composite. Note that it isn't necessary to know the exact value of the hidden number. In this test, the hidden number is 30.

solution	system
58	
	no
59	
	yes
78	
	no
78	
	no
2	
	no
prime	

59 is a divisor of the hidden number. In the interval $[2, 100]$ there is only one number with this divisor. The hidden number must be 59, which is prime. Note that the answer is known even after the second query and you could print it then and terminate. Though, it isn't forbidden to ask unnecessary queries (unless you exceed the limit of 20 queries).

E. BerSU Ball

1 second, 256 megabytes

The Berland State University is hosting a ballroom dance in celebration of its 100500-th anniversary! n boys and m girls are already busy rehearsing waltz, minuet, polonaise and quadrille moves.

We know that several boy&girl pairs are going to be invited to the ball. However, the partners' dancing skill in each pair must differ by at most one.

For each boy, we know his dancing skills. Similarly, for each girl we know her dancing skills. Write a code that can determine the largest possible number of pairs that can be formed from n boys and m girls.

Input

The first line contains an integer n $(1 \le n \le 100)$ — the number of boys. The second line contains sequence $a_1, a_2, ..., a_n$ $(1 \le a_i \le 100)$, where a_i is the i-th boy's dancing skill.

Similarly, the third line contains an integer m $(1 \le m \le 100)$ — the number of girls. The fourth line contains sequence $b_1, b_2, ..., b_m$ $(1 \le b_j \le 100)$, where b_j is the j-th girl's dancing skill.

Output

Print a single number — the required maximum possible number of pairs.

input
```
4
1 4 6 2
5
5 1 5 7 9
```

output

```
3
```

input

```
4
1 2 3 4
4
10 11 12 13
```

output

```
0
```

input

```
5
1 1 1 1 1
3
1 2 3
```

output

```
2
```

F. The Labyrinth

1 second, 256 megabytes

You are given a rectangular field of $n \times m$ cells. Each cell is either empty or impassable (contains an obstacle). Empty cells are marked with '`.`', impassable cells are marked with '`*`'. Let's call two empty cells `adjacent` if they share a side.

Let's call a *connected component* any non-extendible set of cells such that any two of them are connected by the path of adjacent cells. It is a typical well-known definition of a connected component.

For each impassable cell (x, y) imagine that it is an empty cell (all other cells remain unchanged) and find the size (the number of cells) of the connected component which contains (x, y). You should do it for each impassable cell independently.

The answer should be printed as a matrix with n rows and m columns. The j-th symbol of the i-th row should be "`.`" if the cell is empty at the start. Otherwise the j-th symbol of the i-th row should contain the only digit —- the answer modulo 10. The matrix should be printed without any spaces.

To make your output faster it is recommended to build the output as an array of n strings having length m and print it as a sequence of lines. It will be much faster than writing character-by-character.

As input/output can reach huge size it is recommended to use fast input/output methods: for example, prefer to use `scanf/printf` instead of `cin/cout` in C++, prefer to use `BufferedReader/PrintWriter` instead of `Scanner/System.out` in `Java`.

Input

The first line contains two integers n, m $(1 \le n, m \le 1000)$ — the number of rows and columns in the field.

Each of the next n lines contains m symbols: "`.`" for empty cells, "`*`" for impassable cells.

Output

Print the answer as a matrix as described above. See the examples to precise the format of the output.

input

```
3 3
*.*
.*.
*.*
```

output

```
3.3
.5.
3.3
```

input

```
4 5
**..*
..***
.*.*.
*.*.*
```

output

```
46..3
..732
.6.4.
5.4.3
```

In first example, if we imagine that the central cell is empty then it will be included to component of size 5 (cross). If any of the corner cell will be empty then it will be included to component of size 3 (corner).

G. Cards

2 seconds, 256 megabytes

Catherine has a deck of n cards, each of which is either red, green, or blue. As long as there are at least two cards left, she can do one of two actions:

- take any two (not necessarily adjacent) cards with different colors and exchange them for a new card of the third color;
- take any two (not necessarily adjacent) cards with the same color and exchange them for a new card with that color.

She repeats this process until there is only one card left. What are the possible colors for the final card?

Input
The first line of the input contains a single integer n $(1 \le n \le 200)$ — the total number of cards.

The next line contains a string s of length n — the colors of the cards. s contains only the characters 'B', 'G', and 'R', representing blue, green, and red, respectively.

Output
Print a single string of up to three characters — the possible colors of the final card (using the same symbols as the input) in alphabetical order.

input
```
2
RB
```
output
```
G
```

input
```
3
GRG
```
output
```
BR
```

input
```
5
BBBBB
```
output
```
B
```

In the first sample, Catherine has one red card and one blue card, which she must exchange for a green card.

In the second sample, Catherine has two green cards and one red card. She has two options: she can exchange the two green cards for a green card, then exchange the new green card and the red card for a blue card. Alternatively, she can exchange a green and a red card for a blue card, then exchange the blue card and remaining green card for a red card.

In the third sample, Catherine only has blue cards, so she can only exchange them for more blue cards.

H. Partition

1 second, 256 megabytes

You are given a sequence a consisting of n integers. You may partition this sequence into two sequences b and c in such a way that every element belongs exactly to one of these sequences.

Let B be the sum of elements belonging to b, and C be the sum of elements belonging to c (if some of these sequences is empty, then its sum is 0). What is the maximum possible value of $B - C$?

Input
The first line contains one integer n $(1 \le n \le 100)$ — the number of elements in a.

The second line contains n integers $a_1, a_2, ..., a_n$ $(-100 \le a_i \le 100)$ — the elements of sequence a.

Output
Print the maximum possible value of $B - C$, where B is the sum of elements of sequence b, and C is the sum of elements of sequence c.

input
```
3
1 -2 0
```
output
```
3
```

input

```
6
16 23 16 15 42 8
```

output

```
120
```

In the first example we may choose $b = \{1, 0\}$, $c = \{-2\}$. Then $B = 1$, $C = -2$, $B - C = 3$.

In the second example we choose $b = \{16, 23, 16, 15, 42, 8\}$, $c = \{\}$ (an empty sequence). Then $B = 120$, $C = 0$, $B - C = 120$.

I. Trip For Meal

1 second, 512 megabytes

Winnie-the-Pooh likes honey very much! That is why he decided to visit his friends. Winnie has got three best friends: Rabbit, Owl and Eeyore, each of them lives in his own house. There are winding paths between each pair of houses. The length of a path between Rabbit's and Owl's houses is a meters, between Rabbit's and Eeyore's house is b meters, between Owl's and Eeyore's house is c meters.

For enjoying his life and singing merry songs Winnie-the-Pooh should have a meal n times a day. Now he is in the Rabbit's house and has a meal for the first time. Each time when in the friend's house where Winnie is now the supply of honey is about to end, Winnie leaves that house. If Winnie has not had a meal the required amount of times, he comes out from the house and goes to someone else of his two friends. For this he chooses one of two adjacent paths, arrives to the house on the other end and visits his friend. You may assume that when Winnie is eating in one of his friend's house, the supply of honey in other friend's houses recover (most probably, they go to the supply store).

Winnie-the-Pooh does not like physical activity. He wants to have a meal n times, traveling minimum possible distance. Help him to find this distance.

Input

First line contains an integer n $(1 \le n \le 100)$ — number of visits.

Second line contains an integer a $(1 \le a \le 100)$ — distance between Rabbit's and Owl's houses.

Third line contains an integer b $(1 \le b \le 100)$ — distance between Rabbit's and Eeyore's houses.

Fourth line contains an integer c $(1 \le c \le 100)$ — distance between Owl's and Eeyore's houses.

Output

Output one number — minimum distance in meters Winnie must go through to have a meal n times.

input

```
3
2
3
1
```

output

```
3
```

input

```
1
2
3
5
```

output

```
0
```

In the first test case the optimal path for Winnie is the following: first have a meal in Rabbit's house, then in Owl's house, then in Eeyore's house. Thus he will pass the distance $2 + 1 = 3$.

In the second test case Winnie has a meal in Rabbit's house and that is for him. So he doesn't have to walk anywhere at all.

J. Valera and Antique Items

1 second, 256 megabytes

Valera is a collector. Once he wanted to expand his collection with exactly one antique item.

Valera knows n sellers of antiques, the i-th of them auctioned k_i items. Currently the auction price of the j-th object of the i-th seller is s_{ij}. Valera gets on well with each of the n sellers. He is perfectly sure that if he outbids the current price of one of the items in the auction (in other words, offers the seller the money that is strictly greater than the current price of the item at the auction), the seller of the object will immediately sign a contract with him.

Unfortunately, Valera has only v units of money. Help him to determine which of the n sellers he can make a deal with.

Input

The first line contains two space-separated integers n, v $(1 \le n \le 50;\ 10^4 \le v \le 10^6)$ — the number of sellers and the units of money the Valera has.

Then n lines follow. The i-th line first contains integer k_i $(1 \le k_i \le 50)$ the number of items of the i-th seller. Then go k_i space-separated integers $s_{i1}, s_{i2}, ..., s_{ik_i}$ $(10^4 \le s_{ij} \le 10^6)$ — the current prices of the items of the i-th seller.

Output

In the first line, print integer p — the number of sellers with who Valera can make a deal.

In the second line print p space-separated integers $q_1, q_2, ..., q_p$ $(1 \le q_i \le n)$ — the numbers of the sellers with who Valera can make a deal. Print the numbers of the sellers **in the increasing order**.

input

```
3 50000
1 40000
2 20000 60000
3 10000 70000 190000
```

output

```
3
1 2 3
```

input

```
3 50000
1 50000
3 100000 120000 110000
3 120000 110000 120000
```

output

```
0
```

In the first sample Valera can bargain with each of the sellers. He can outbid the following items: a 40000 item from the first seller, a 20000 item from the second seller, and a 10000 item from the third seller.

In the second sample Valera can not make a deal with any of the sellers, as the prices of all items in the auction too big for him.

K. Winter Is Coming

1 second, 256 megabytes

The winter in Berland lasts n days. For each day we know the forecast for the average air temperature that day.

Vasya has a new set of winter tires which allows him to drive safely no more than k days at any average air temperature. After k days of using it (regardless of the temperature of these days) the set of winter tires wears down and cannot be used more. It is not necessary that these k days form a continuous segment of days.

Before the first winter day Vasya still uses **summer tires**. It is possible to drive safely on summer tires any number of days when the average air temperature is **non-negative**. It is impossible to drive on summer tires at days when the average air temperature is negative.

Vasya can change summer tires to winter tires and vice versa at the beginning of any day.

Find the minimum number of times Vasya needs to change summer tires to winter tires and vice versa to drive safely during the winter. At the end of the winter the car can be with any set of tires.

Input

The first line contains two positive integers n and k $(1 \le n \le 2 \cdot 10^5, 0 \le k \le n)$ — the number of winter days and the number of days winter tires can be used. It is allowed to drive on winter tires at any temperature, but no more than k days in total.

The second line contains a sequence of n integers $t_1, t_2, ..., t_n$ $(-20 \le t_i \le 20)$ — the average air temperature in the i-th winter day.

Output

Print the minimum number of times Vasya has to change summer tires to winter tires and vice versa to drive safely during all winter. If it is impossible, print `-1`.

input

```
4 3
-5 20 -3 0
```

output

```
2
```

input

```
4 2
-5 20 -3 0
```

output

```
4
```

input

```
10 6
2 -5 1 3 0 0 -4 -3 1 0
```

output

```
3
```

In the first example before the first winter day Vasya should change summer tires to winter tires, use it for three days, and then change winter tires to summer tires because he can drive safely with the winter tires for just three days. Thus, the total number of tires' changes equals two.

In the second example before the first winter day Vasya should change summer tires to winter tires, and then after the first winter day change winter tires to summer tires. After the second day it is necessary to change summer tires to winter tires again, and after the third day it is necessary to change winter tires to summer tires. Thus, the total number of tires' changes equals four.

L. Leaving Auction

2 seconds, 256 megabytes

There are n people taking part in auction today. The rules of auction are classical. There were n bids made, though it's not guaranteed they were from different people. It might happen that some people made no bids at all.

Each bid is define by two integers (a_i, b_i), where a_i is the index of the person, who made this bid and b_i is its size. Bids are given in chronological order, meaning $b_i < b_{i+1}$ for all $i < n$. Moreover, participant never makes two bids in a row (no one updates his own bid), i.e. $a_i \neq a_{i+1}$ for all $i < n$.

Now you are curious with the following question: who (and which bid) will win the auction if some participants were absent? Consider that if someone was absent, all his bids are just removed and no new bids are added.

Note, that if during this imaginary exclusion of some participants it happens that some of the remaining participants makes a bid twice (or more times) in a row, only first of these bids is counted. For better understanding take a look at the samples.

You have several questions in your mind, compute the answer for each of them.

Input

The first line of the input contains an integer n $(1 \leq n \leq 200\,000)$ — the number of participants and bids.

Each of the following n lines contains two integers a_i and b_i $(1 \leq a_i \leq n, 1 \leq b_i \leq 10^9, b_i < b_{i+1})$ — the number of participant who made the i-th bid and the size of this bid.

Next line contains an integer q $(1 \leq q \leq 200\,000)$ — the number of question you have in mind.

Each of next q lines contains an integer k $(1 \leq k \leq n)$, followed by k integers l_j $(1 \leq l_j \leq n)$ — the number of people who are not coming in this question and their indices. It is guarenteed that l_j values are different for a single question.

It's guaranteed that the sum of k over all question won't exceed $200\,000$.

Output

For each question print two integer — the index of the winner and the size of the winning bid. If there is no winner (there are no remaining bids at all), print two zeroes.

input

```
6
1 10
2 100
3 1000
1 10000
2 100000
3 1000000
3
1 3
2 2 3
2 1 2
```

output

```
2 100000
1 10
3 1000
```

input

```
3
1 10
2 100
1 1000
2
2 1 2
2 2 3
```

output

```
0 0
1 10
```

Consider the first sample:

- In the first question participant number 3 is absent so the sequence of bids looks as follows:

  1. $1\ 10$
  2. $2\ 100$
  3. $1\ 10\,000$
  4. $2\ 100\,000$

  Participant number 2 wins with the bid $100\,000$.
- In the second question participants 2 and 3 are absent, so the sequence of bids looks:

  1. $1\ 10$
  2. $1\ 10\,000$

  The winner is, of course, participant number 1 but the winning bid is 10 instead of $10\,000$ as no one will ever increase his own bid (in this problem).
- In the third question participants 1 and 2 are absent and the sequence is:

  1. $3\ 1\,000$
  2. $3\ 1\,000\,000$

  The winner is participant 3 with the bid $1\,000$.

M. Mishka and Game

1 second, 256 megabytes

Mishka is a little polar bear. As known, little bears loves spending their free time playing dice for chocolates. Once in a wonderful sunny morning, walking around blocks of ice, Mishka met her friend Chris, and they started playing the game.

Rules of the game are very simple: at first number of rounds n is defined. In every round each of the players throws a cubical dice with distinct numbers from 1 to 6 written on its faces. Player, whose value after throwing the dice is greater, wins the round. In case if player dice values are equal, no one of them is a winner.

In average, player, who won most of the rounds, is the winner of the game. In case if two players won the same number of rounds, the result of the game is draw.

Mishka is still very little and can't count wins and losses, so she asked you to watch their game and determine its result. Please help her!

Input

The first line of the input contains single integer n n $(1 \le n \le 100)$ — the number of game rounds.

The next n lines contains rounds description. i-th of them contains pair of integers m_i and c_i $(1 \le m_i,\ c_i \le 6)$ — values on dice upper face after Mishka's and Chris' throws in i-th round respectively.

Output

If Mishka is the winner of the game, print `"Mishka"` (without quotes) in the only line.

If Chris is the winner of the game, print `"Chris"` (without quotes) in the only line.

If the result of the game is draw, print `"Friendship is magic!^^"` (without quotes) in the only line.

input
```
3
3 5
2 1
4 2
```
output
```
Mishka
```

input
```
2
6 1
1 6
```
output
```
Friendship is magic!^^
```

input
```
3
1 5
3 3
2 2
```

output

```
Chris
```

In the first sample case Mishka loses the first round, but wins second and third rounds and thus she is the winner of the game.

In the second sample case Mishka wins the first round, Chris wins the second round, and the game ends with draw with score 1:1.

In the third sample case Chris wins the first round, but there is no winner of the next two rounds. The winner of the game is Chris.

N. The String Has a Target

1 second, 256 megabytes

You are given a string s. You can apply this operation to the string exactly once: choose index i and move character s_i to the beginning of the string (removing it at the old position). For example, if you apply the operation with index $i = 4$ to the string "`abaacd`" with numbering from 1, you get the string "`aabacd`". What is the lexicographically minimal† string you can obtain by this operation?

† A string a is lexicographically smaller than a string b of the same length if and only if the following holds:

- in the first position where a and b differ, the string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 10^5$) — the length of the string.

The second line of each test case contains the string s of length n, consisting of lowercase English letters.

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Output

For each test case, on a separate line print the lexicographically smallest string that can be obtained after applying the operation to the original string exactly once.

input

```
4
3
cba
4
acac
5
abbcb
4
aaba
```

output

```
acb
aacc
abbcb
aaab
```

In the first test case, you need to move the last character to the beginning.

In the second case, you need to move the second letter "a".

In the third set you need to apply the operation with $i = 1$, then the string will not change.